Exploratory Data Analysis

> This book is about exploratory data analysis, about looking at data to see what it seems to say. It concentrates on simple arithmetic and easy-to-draw pictures. It regards whatever appearances we have recognized as partial descriptions, and tries to look beneath them for new insights. Its concern is with appearance, not with confirmation. @TukeyExploratoryDataAnalysis1977 v

- Exploratory Data Analysis (EDA) is "an attitude, AND a flexibility, AND some graph paper (or transparencies, or both)" [@TukeyWeNeedBoth1980] developed by statistician John Tukey

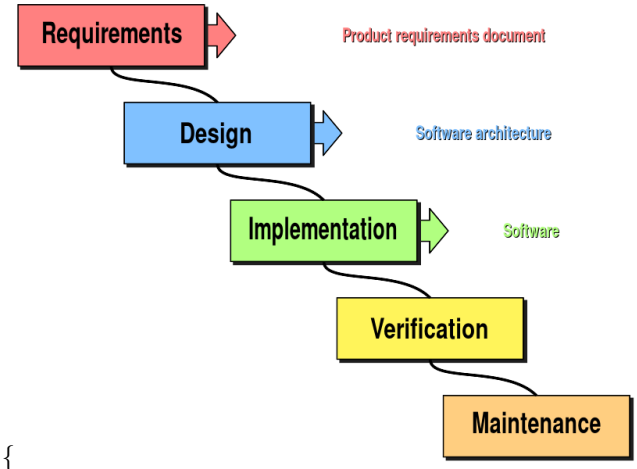
Exploratory and Confirmatory Research

- Especially in the wake of the replication crisis, one common distinction is between exploratory and confirmatory research [@WagenmakersAgendaPurelyConfirmatory2012]

Confirmatory	Exploratory
hypothesis testing	hypothesis development
specified in advance	adaptable
algorithmic	free, creative
avoids inferential errors (type I, false positives)	*makes* errors?
rigorous	**lacking rigor?**
real science??	**h*cking around with data??**

Better Models for EDA I: Risk Management in Spiral Development

\begin{figure}



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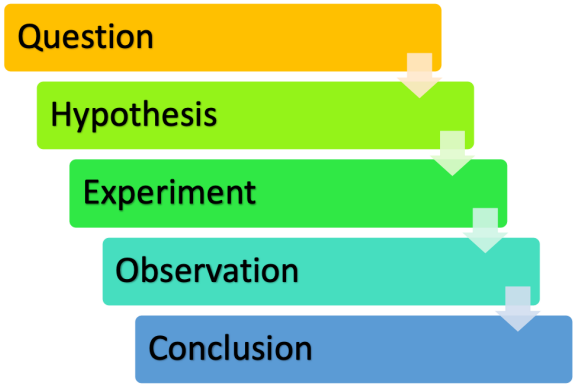

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\caption{The waterfall model of software development resembles the elementary school model of "the scientific method," with an inflexible, linear sequence of predefined steps. Source: https://en.wikipedia.org/wiki/Waterfall_model} \end{figure}

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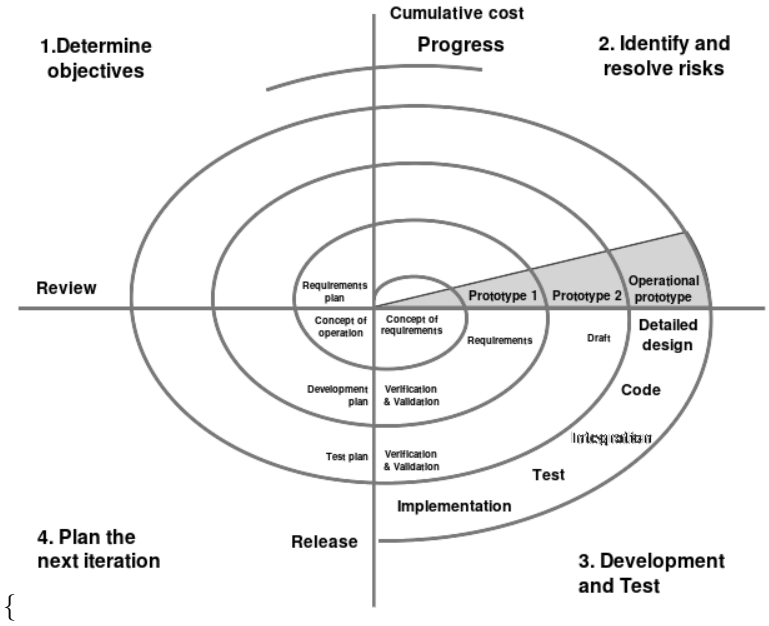

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\caption{Boehm's spiral model of software development, based on @BoehmSpiralModelSoftware1988, fig. 2. Source: https://en.wikipedia.org/wiki/Spiral_model} \end{figure}

- Through the 1970s, software engineering relied on a "waterfall model" of development
  - Resembles elementary school picture of "the scientific method": linear sequences of steps, specified in advance

@BoehmSpiralModelSoftware1988's "spiral model"

- Critique: The waterfall model required extensive advance planning
  - Couldn't be adapted as new problems were identified or requirements changed

- **Basic idea: Software development works in an expanding spiral, moving from low-cost, low-risk proof of concept to usable demonstration to full product**

1. Define outputs concurrently, not sequentially
2. 4 basic activities in each cycle
   a. Identify win conditions for all stakeholders
   b. Identify and evaluate alternative approaches
   c. **Identify and resolve risks for each approach**
   d. Obtain approval and commitment from all stakeholders
3. **Risk determines level of effort**
4. Risk determines degree of details
5. Use anchor point milestones
   a. Is there a sufficient definition of a technical and management approach to satisfying everyone's win conditions?
   b. Is there a sufficient definition of the preferred approach to satisfying everyone's win conditions, and are all significant risks eliminated or mitigated?
   c. Is there sufficient preparation of the software, site, users, operators, and maintainers to satisfy everyone's win conditions by launching the system?

Scientific research projects using the spiral model

- Start small: pilot study, subset of big data
- Stakeholders
  - PI, grad researchers, undergrad researchers
  - Dissertation committee, advisor; tenure committee
  - Journal editors and reviewers, funding agency, hiring committees
  - Community partners? Policymakers? Industry?
- **EDA: Identify and mitigate risks before you fully commit to an approach**
  - Do these data include the main variables we care about?
  - How about additional covariates, predictors, identifiers we need?
  - Where are the data errors? Can we fix them?
  - Where are the missing values?

Better Models for EDA II: Epicycle of Analysis

- @PengArtDataScience2016
  - Data analysis is organized into 5 activities
  - Each activity involves the same 3-step "epicycle" process
    1. Develop expectations
    2. Collect information
    3. Compare and revise expectations
  - Not "the scientific method"! [@PengArtDataScience2016 4]
    - "Highly iterative and non-linear"
    - "information is learned at each step, which then informs
      - **whether (and how) to refine, and redo, the [previous] step** …, or
      - whether (and how) to proceed to the next step."

Table 2: Aligning the goals of EDA with steps in the "epicycle of analysis" [@PengArtDataScience2016]

Goals of EDA	Epicycle step
Determine if there are problems with the data	2. Collecting information
Determine whether our question can be answered with these data	3. Comparing and revising expectations

Goals of EDA	Epicycle step
Develop sketch of an answer	1. Developing expectations

Better Models for EDA III: Phenomena Development

[todo]

- @BogenSavingPhenomena1988, §§ intro and III-IV (303-6, 315-25)
- is "theory" the right term?
  - mechanisms; causal processes; explanations
- note on the stability of phenomena
  - the pattern is present here and now
  - the question is, will it be present elsewhere in the future?
  - replicability
  - as an answer to worries about type I errors, p-hacking

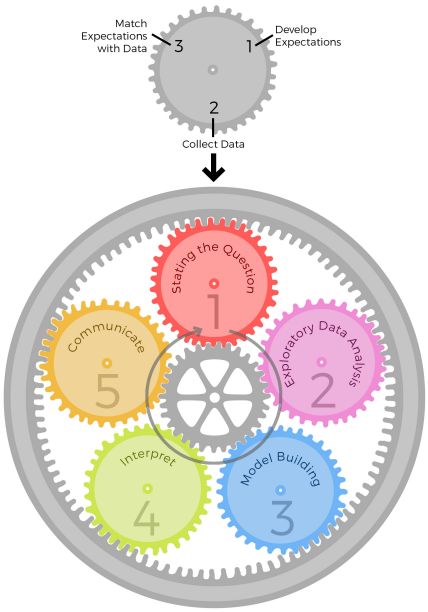

Figure 1: Epicycle of analysis model, @PengArtDataScience2016, 5